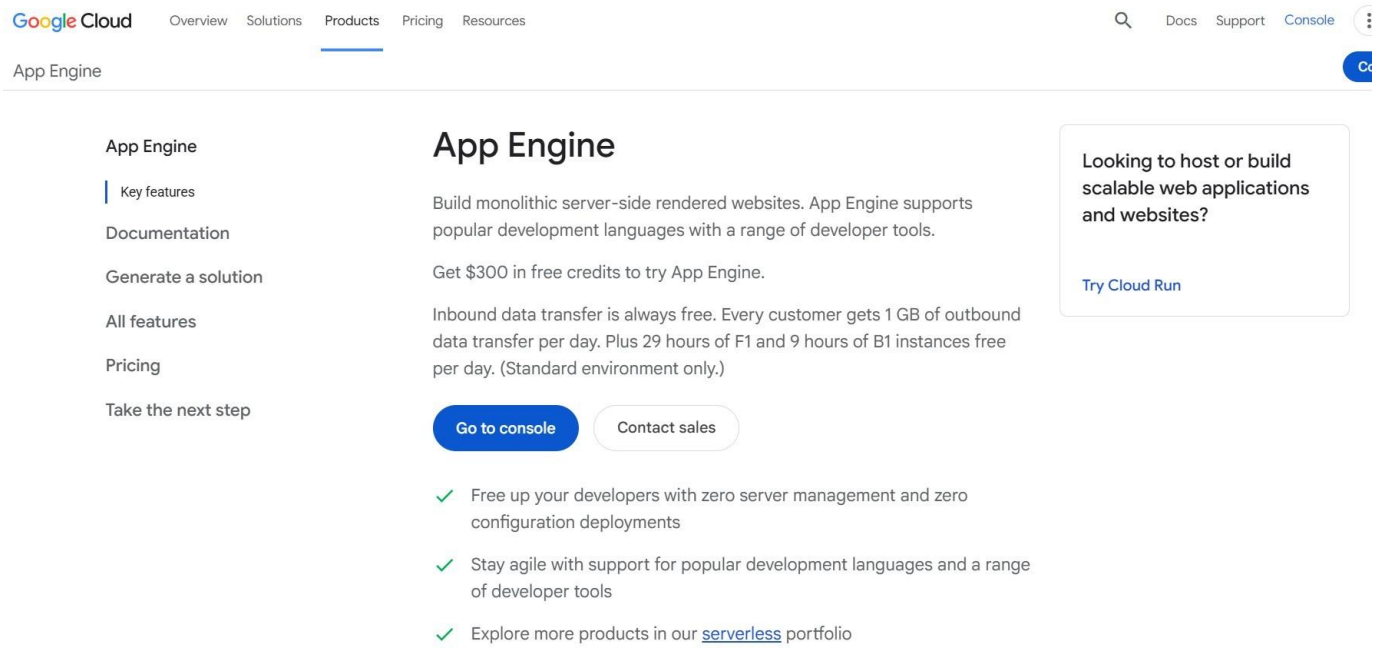
Week-7

Aim: Use GAE launcher to launch the web applications.

Google App Engine (GAE): A platform-as-a-service (PaaS) offering from Google Cloud that allows developers to build, deploy, and scale applications without managing the underlying infrastructure.

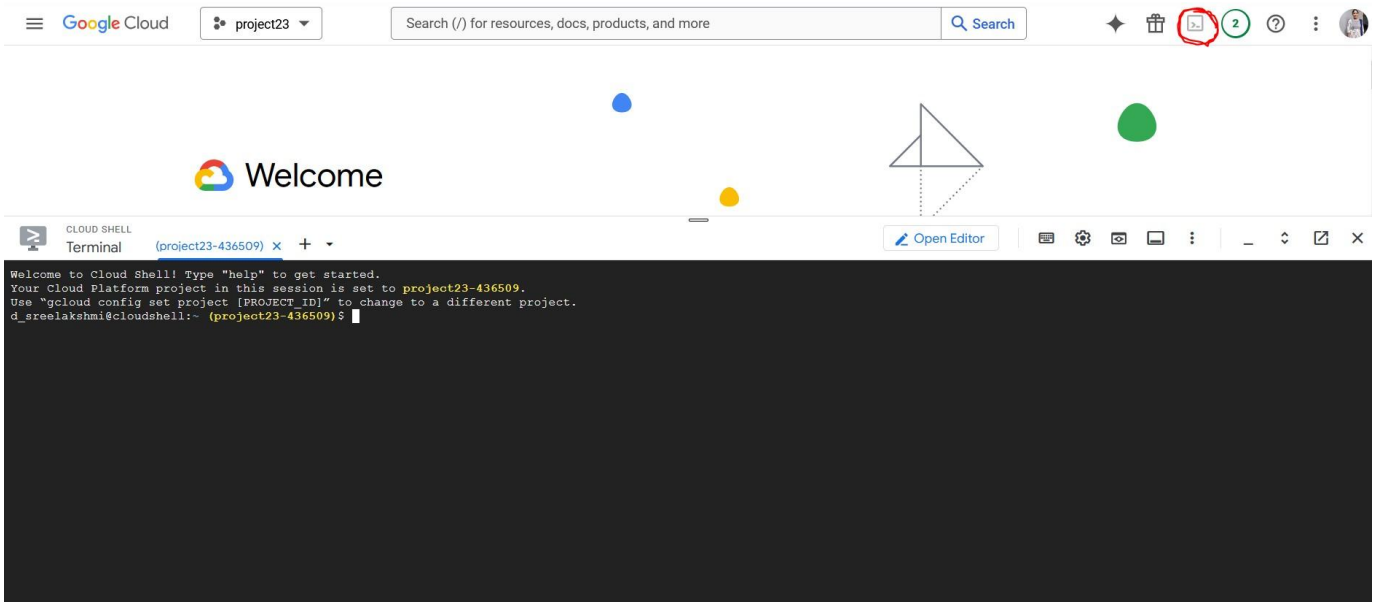
GAE Launcher: Google Cloud Shell is an interactive, browser-based shell environment for managing and interacting with Google Cloud resources. It provides a command-line interface (CLI) that comes preconfigured with essential tools and libraries needed to work with Google Cloud.

1. **Signin to Google App Engine:** click on console



2. **Open Google Cloud Shell:**

   - Go to the Google Cloud Console.
   - Click the Cloud Shell icon in the top right corner. This will open a terminal in your browser with all the tools pre-installed.



3. Create a New Directory:

- In the Cloud Shell terminal, create a new directory for your web app **mkdir Roll no**
- Change to the new directory **mkdir Roll no**

4: Set Up Your Google Cloud Project

1. Authenticate with Google Cloud: gcloud auth login
2. Create a new project: gcloud projects create cc-week-7
3. Verify the list of projects: gcloud projects list
4. Set the project ID in the CLI:gcloud config set project cc-week-7
5. Describe the project: gcloud projects describe cc-week-6

5. Create a Simple Python Web Application:

nano app.py

```
from flask import Flask,
render_template app =
Flask(__name__) @app.route('/') def
home():
   return
render_template('webapp.html') if
__name__ == '__main__':
app.run(debug=True)
```

Create the webapp.html in a templates folder

Place your webapp.html in a folder named templates in the same directory as app.py. The directory structure should look like this:

```
/your_project_directory
   /templates
webapp.html     app.py
```

nano webapp.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Hello World</title>
</head>
<body>
<p>Hello World!</p>
</body>
```

```
</html>
```

4. Create the App Engine Configuration File

nano app.yaml

```
runtime: python39
handlers:
 - url: /.*
script: auto
```

This configuration tells Google App Engine to use Python 3.9 and run the application.

5. Deploy the Applicationgcloud

app deploy

6. To View your application gcloud

app browse python app.py

OUTPUT:

Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`. g24251a1261@cloudshell:~$ gcloud auth login

You are already authenticated with gcloud when running inside the Cloud Shell and so do not need to run this command. Do you wish to proceed anyway?

Do you want to continue (Y/n)? y

Go to the following link in your browser, and complete the sign-in prompts:

https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcode.html&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=4zqsAQeMQgxPN3dMZItLJ2HS38aa4e&prompt=consent&token_usage=remote&access_type=offline&code_challenge=og01bZyiGAh9fyIakSbwo7uPGac72JGf0OkwQ1AguSg&code_challenge_method=S256

Once finished, enter the verification code provided in your browser:
4/0ATsMZqDBE7tbMcrC10evwBuC76ER7sEjSoErNO_NQzG-i-eu-At_xP1lF4wm4_UGFF-0Eg

You are now logged in as [24251a1261@gmail.com].
Your current project is [None]. You can change this setting by running:
$ gcloud config set project PROJECT_ID g24251a1261@cloudshell:~$ gcloud projects list
PROJECT_ID: cc-week-6-1261
NAME: cc-week-6-1261
PROJECT_NUMBER: 419102498409 ENVIRONMENT:

PROJECT_ID: ccexam1261
NAME: ccexam1261
PROJECT_NUMBER: 411770166941 ENVIRONMENT:
g24251a1261@cloudshell:~$ gcloud projects create cc-week7-1261
Create in progress for [https://cloudresourcemanager.googleapis.com/v1/projects/cc-week7-1261].
Waiting for [operations/create_project.global.6791099454480495478] to finish...done.
Enabling service [cloudapis.googleapis.com] on project [cc-week7-1261]...
Operation "operations/acat.p2-846525606862-65a6edc7-71cc-470f-b9f0-f9c7331cd9c5" finished successfully.
g24251a1261@cloudshell:~$ gcloud config set project cc-week7-1261 Updated property [core/project].
g24251a1261@cloudshell:~ (cc-week7-1261)$ cd %USERPROFILE%
-bash: cd: %USERPROFILE%: No such file or directory g24251a1261@cloudshell:~ (cc-week7-1261)$ MKDIR 24251A1261

-bash: MKDIR: command not found
g24251a1261@cloudshell:~ (cc-week7-1261)$ mkdir 24251A1261 g24251a1261@cloudshell:~ (cc-week7-25)$ cd 24251A1261 g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ nano app.py g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ nano app.yaml
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ nano requirements.txt
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ mkdir templates
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ cd templates
g24251a1261@cloudshell:~/24251A1261/templates (cc-week7-1261)$ nano webapp.html
g24251a1261@cloudshell:~/24251A1261/templates (cc-week7-1261)$ cd ..
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ nano app.py
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead. * Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (inotify) * Debugger is active!
* Debugger PIN: 132-184-982
^Cg24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ nano app.yaml
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead. * Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (inotify) * Debugger is active!
* Debugger PIN: 132-184-982
^Cg24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ nano app.yaml
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ cd templates
g24251a1261@cloudshell:~/24251A1261/templates (cc-week7-1261)$ nano webapp.html
g24251a1261@cloudshell:~/24251A1261/templates (cc-week7-1261)$ cd ..
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead. * Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (inotify) * Debugger is active!
* Debugger PIN: 132-184-982
127.0.0.1 - - [09/Sep/2026 09:17:16] "GET /?authuser=0 HTTP/1.1" 200 -
127.0.0.1 - - [09/Sep/2026 09:17:16] "GET /favicon.ico HTTP/1.1" 404
^Cg24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ gcloud init Welcome!
This command will take you through the configuration of gcloud.

Settings from your current configuration [cloudshell-5124] are:
accessibility: screen_reader: 'True' component_manager:
disable_update_check: 'True' compute:
gce_metadata_read_timeout_sec: '30' core:
account: 24251a1261@gmail.com
disable_usage_reporting: 'False'
project: cc-week7-1261
universe_domain: googleapis.com metrics:
environment: devshell

Pick configuration to use:
[1] Re-initialize this configuration [cloudshell-5124] with new settings
[2] Create a new configuration
Please enter your numeric choice: 1

Your current configuration has been set to: [cloudshell-5124]

You can skip diagnostics next time by using the following flag: gcloud init
--skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).

Choose the account you want to use for this configuration.
To use a federated user account, exit this command and sign in to the gcloud CLI with your login configuration file, then run this command again.

Select an account:
[1] 24251a1261@gmail.com
[2] Sign in with a new Google Account
[3] Skip this step
Please enter your numeric choice: 1

You are signed in as: [24251a1261@gmail.com].

Pick cloud project to use:
[1] cc-week-6-1261
[2] cc-week7-1261
[3] ccexam1261
[4] Enter a project ID
[5] Create a new project
Please enter numeric choice or text value (must exactly match list item): 2

Your current project has been set to: [cc-week7-1261].

Not setting default zone/region (this feature makes it easier to use [gcloud compute] by setting an appropriate default value for the --zone and --region flag).
See https://cloud.google.com/compute/docs/gcloud-compute section on how to set default compute region and zone manually. If you would like [gcloud init] to be able to do this for you the next time you run it, make sure the Compute Engine API is enabled for your project on the https://console.developers.google.com/apis page.

Created a default .boto configuration file at [/home/g24251a1261/.boto]. See this file and [https://cloud.google.com/storage/docs/gsutil/commands/config] for more information about configuring Google Cloud Storage. The Google Cloud CLI is configured and ready to use!

* Commands that require authentication will use 24251a1261@gmail.com by default
* Commands will reference project `cc-week7-1261` by default
Run `gcloud help config` to learn how to change individual settings

This gcloud configuration is called [cloudshell-5124]. You can create additional configurations if you work with multiple accounts and/or projects. Run `gcloud topic configurations` to learn more.

Some things to try next:

* Run `gcloud --help` to see the Cloud Platform services you can interact with. And run `gcloud help COMMAND` to get help on any gcloud command.
* Run `gcloud topic --help` to learn about advanced features of the CLI like arg files and output formatting
* Run `gcloud cheat-sheet` to see a roster of go-to `gcloud` commands.
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ gcloud app deploy
WARNING: You might be using automatic scaling for a standard environment deployment, without providing a value for automatic_scaling.max_instances. Starting from March, 2025, App Engine sets the automatic scaling maximum instances default for standard environment deployments to 20. This change doesn't impact existing apps. To override the default, specify the new max_instances value in your app.yaml file, and deploy a new version or redeploy over an existing version. For details on max_instances, see https://cloud.google.com/appengine/docs/standard/reference/app-yaml.md#scaling_elements.

You are creating an app for project [cc-week7-1261].
WARNING: Creating an App Engine application for a project is irreversible and the region cannot be changed. More information about regions is at <https://cloud.google.com/appengine/docs/locations>.

WARNING: Starting from March, 2025, App Engine sets the automatic scaling maximum instances default for standard environment deployments to 20. This change doesn't impact

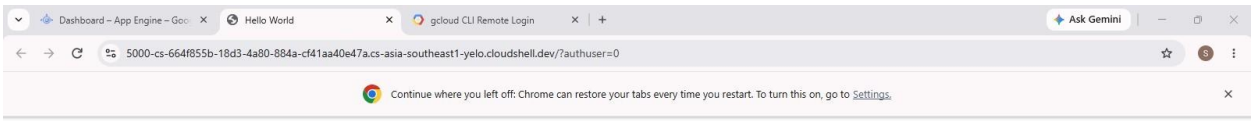
existing apps. To override the default, specify the new max_instances value in your app.yaml file, and deploy a new version or redeploy over an existing version.
For more details on max_instances, see
<https://cloud.google.com/appengine/docs/standard/reference/appyaml.md#scaling_elements>.

NOTE: Cloud Run offers the most modern fully managed application hosting experience with lower minimum billable times and support for GPUs on demand for your AI/ML workloads. Deploy code written in any programming language supported by App Engine on Cloud Run. Learn more at https://cloud.google.com/run/docs/quickstarts#build-and-deploy-a-web-service

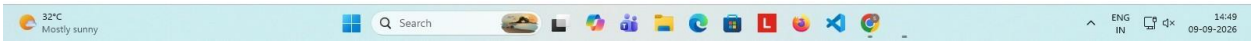
Please choose the region where you want your App Engine application located:

[1] asia-east1 (supports standard and flexible)
[2] asia-east2 (supports standard and flexible and search_api)
[3] asia-northeast1 (supports standard and flexible and search_api) [4] asia-northeast2 (supports standard and flexible and search_api)
[5] asia-northeast3 (supports standard and flexible and search_api)
[6] asia-south1 (supports standard and flexible and search_api)
[7] asia-southeast1 (supports standard and flexible)
[8] asia-southeast2 (supports standard and flexible and search_api)
[9] australia-southeast1 (supports standard and flexible and search_api)
[10] europe-central2 (supports standard and flexible)
[11] europe-west (supports standard and flexible and search_api)
[12] europe-west2 (supports standard and flexible and search_api) [13] europe-west3 (supports standard and flexible and search_api) [14] europe-west6 (supports standard and flexible and search_api)
[15] northamerica-northeast1 (supports standard and flexible and search_api)
[16] southamerica-east1 (supports standard and flexible and search_api)
[17] us-central (supports standard and flexible and search_api) [18] us-east1 (supports standard and flexible and search_api)
[19] us-east4 (supports standard and flexible and search_api)
[20] us-west1 (supports standard and flexible)
[21] us-west2 (supports standard and flexible and search_api)
[22] us-west3 (supports standard and flexible and search_api)
[23] us-west4 (supports standard and flexible and search_api)
[24] cancel
Please enter your numeric choice: 1

ERROR: (gcloud.app.deploy) INVALID_ARGUMENT: The project must have a billing account attached.
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$ gcloud app browse
ERROR: (gcloud.app.browse) The current Google Cloud project [cc-week7-1261] does not contain an App Engine application. Use `gcloud app create` to initialize an App Engine application within the project.
g24251a1261@cloudshell:~/24251A1261 (cc-week7-1261)$


Dashboard – App Engine – Goo
Hello World
gcloud CLI Remote Login
Ask Gemini
5000-cs-664f855b-18d3-4a80-884a-cf41aa40e47a.cs-asia-southeast1-yelo.cloudshell.dev/?authuser=0
Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to Settings.

Hello, World!


32°C
Mostly sunny
Search
ENG
IN
14:49
09-09-2026

Week-8

Aim: Simulate a cloud scenario using CloudSim and run a scheduling algorithm in CloudSim.

Simulation:

- Simulation is making a virtual model of a real system to see how it would act in different situations.
- It helps us test and understand systems safely without changing the real thing.

CloudSim:

- CloudSim is an open-source framework, which is used to simulate cloud computing infrastructure and services.
- It is developed by the CLOUDS Lab organization and is written entirely in Java.
- It is used for modelling and simulating a cloud computing environment

Some of the most common classes used during simulation are:

- **Datacenter:** used for modelling the foundational hardware equipment of any cloud environment, that is the Datacenter.

- **Host:** this class executes actions related to management of virtual machines. It also defines policies for provisioning memory and bandwidth to the virtual machines, as well as allocating CPU cores to the virtual machines.

- **VM:** this class represents a virtual machine by providing data members defining a VM's bandwidth, RAM, mips (million instructions per second), size while also providing setter and getter methods for these parameters.

- **Cloudlet:** a cloudlet class represents any task that is run on a VM, like a processing task, or a memory access task, or a file updating task etc.

- **CloudSim:** this is the class responsible for initializing and starting the simulation environment after all the necessary cloud entities have been defined and later stopping after all the entities have been destroyed.

Benefits of CloudSim:

- Open source and free of cost
- Easy to use and Scalable.
- Risks can be evaluated at an earlier stage.
- Provides pre-defined simulation packages

Scheduling Algorithm:

In CloudSim, a scheduling algorithm schedule how cloud resources, like virtual machines (VMs), are assigned to different tasks or jobs.

Steps:

1. Java Installation

a. Check java in your system

Open Command Prompt (Windows) or Terminal (Mac/Linux).

Type the following command to check the Java version:

java -version

If Java is installed, you'll see the version details.

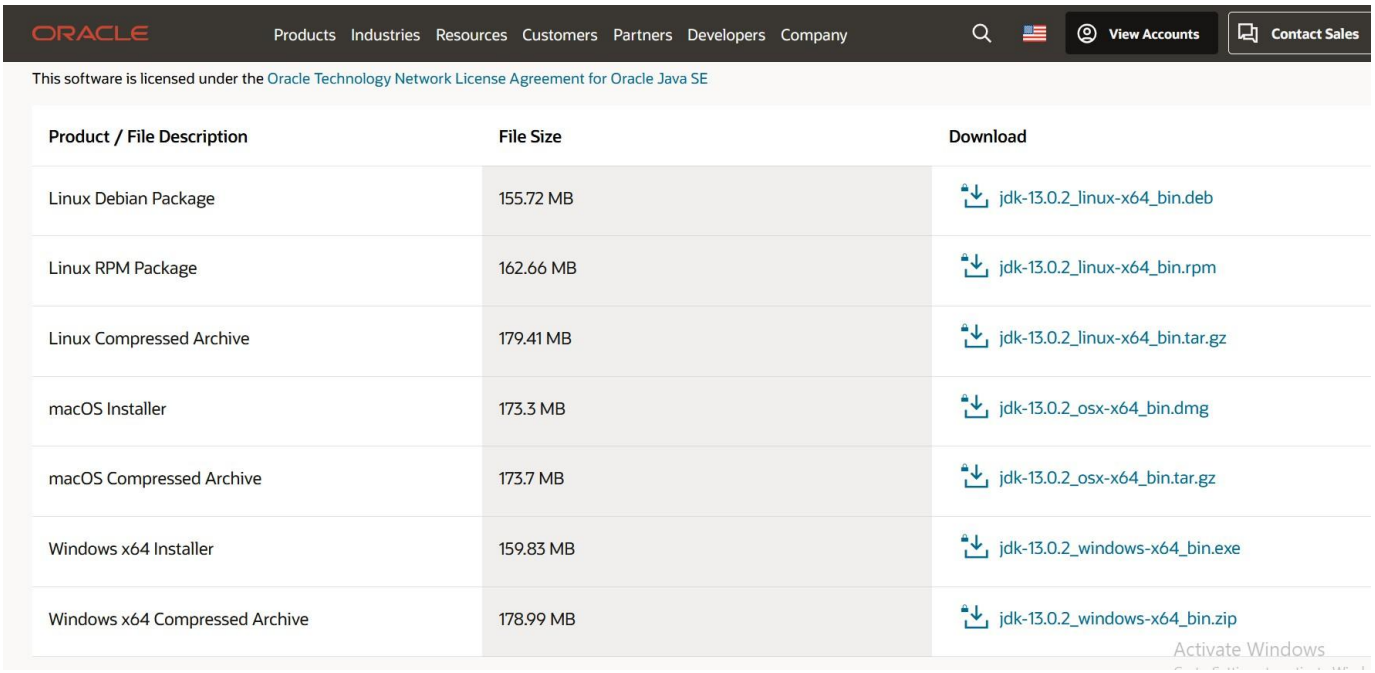
If not installed, proceed to the next step to download and install Java.

Visit the official Java download page.

b. Download the appropriate version for your operating system (64-bit) For Windows, download the **.exe** file.

For Mac, download the **.dmg** file.

For Linux, download the **.tar.gz** file.

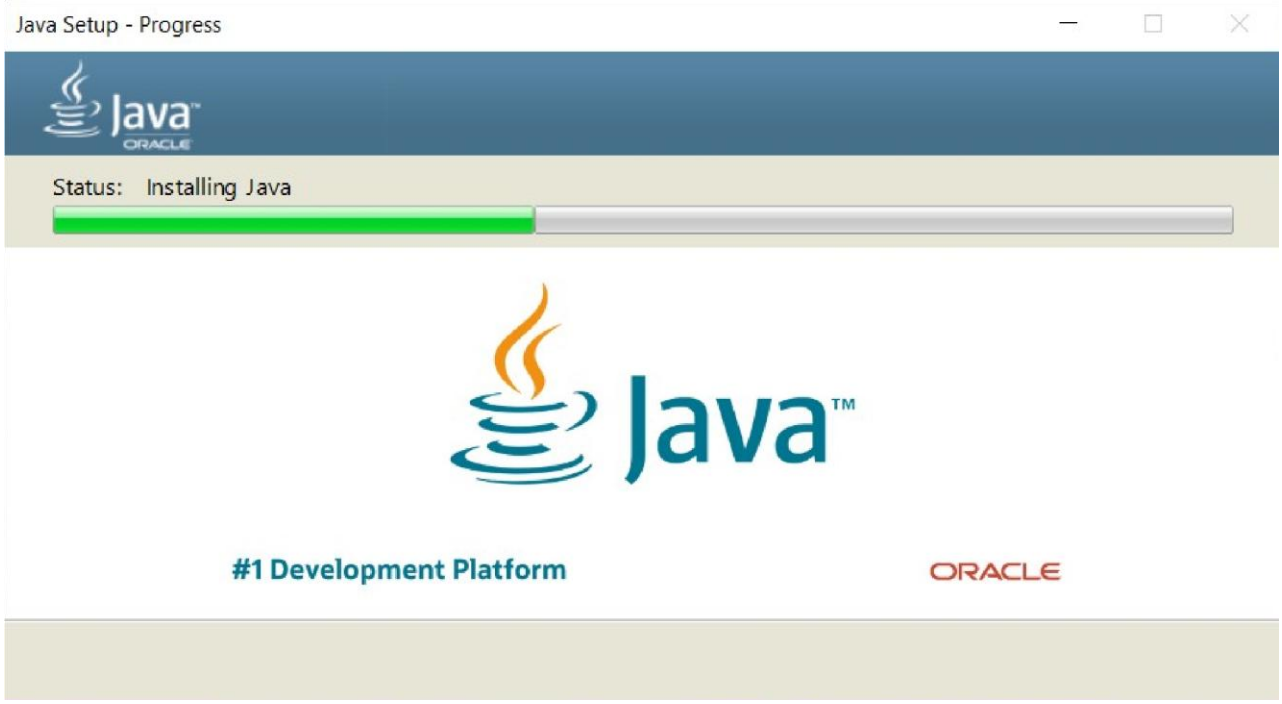


https://www.oracle.com/java/technologies/javase/jdk13-archive-downloads.html

c. Install Java

Run the downloaded setup and follow the installation instructions.

Complete the installation process.



d. Set Path Variable for Java (Windows)

Right-click on This PC or My Computer and go to Properties.

Select advanced system settings.

Click Environment Variables.

In System variables, find Path, select it, and click Edit.

Add the path of the bin folder in the Java installation directory.

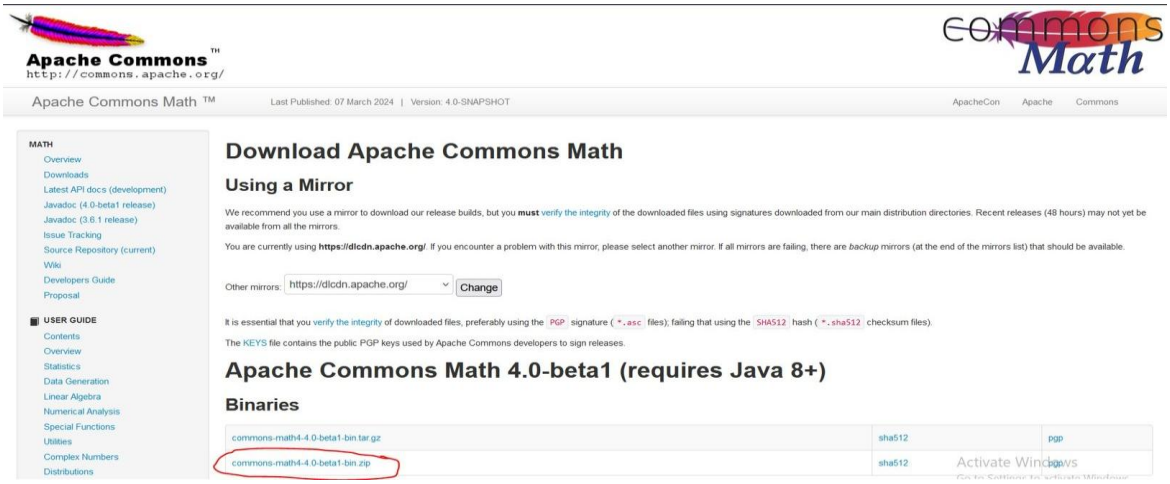
C:\Program Files\Java\jdk-14\bin //Click OK to save the changes.

2. Download CloudSim and additional jar file

a. Download CloudSim

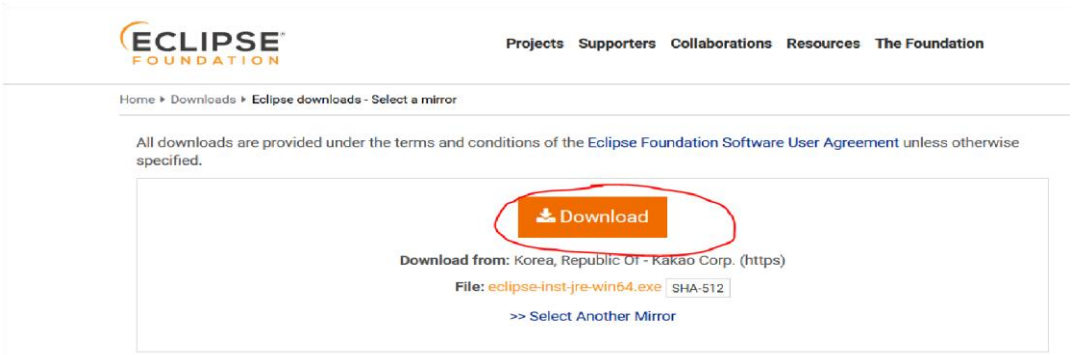
Download CloudSim and extract the contents.

b. Download Commons math 3 jar file

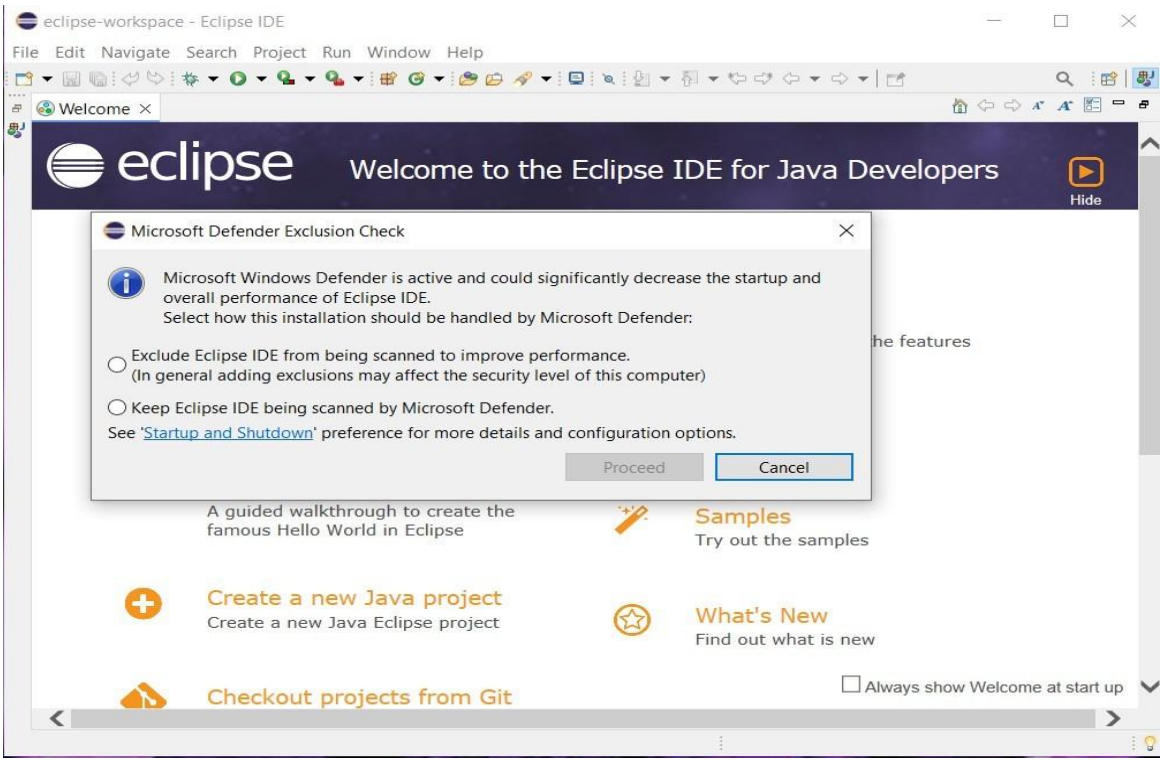


3. Eclipse IDE Installation

➢ If java 64 bit Installed then download 64 bit Eclipse.

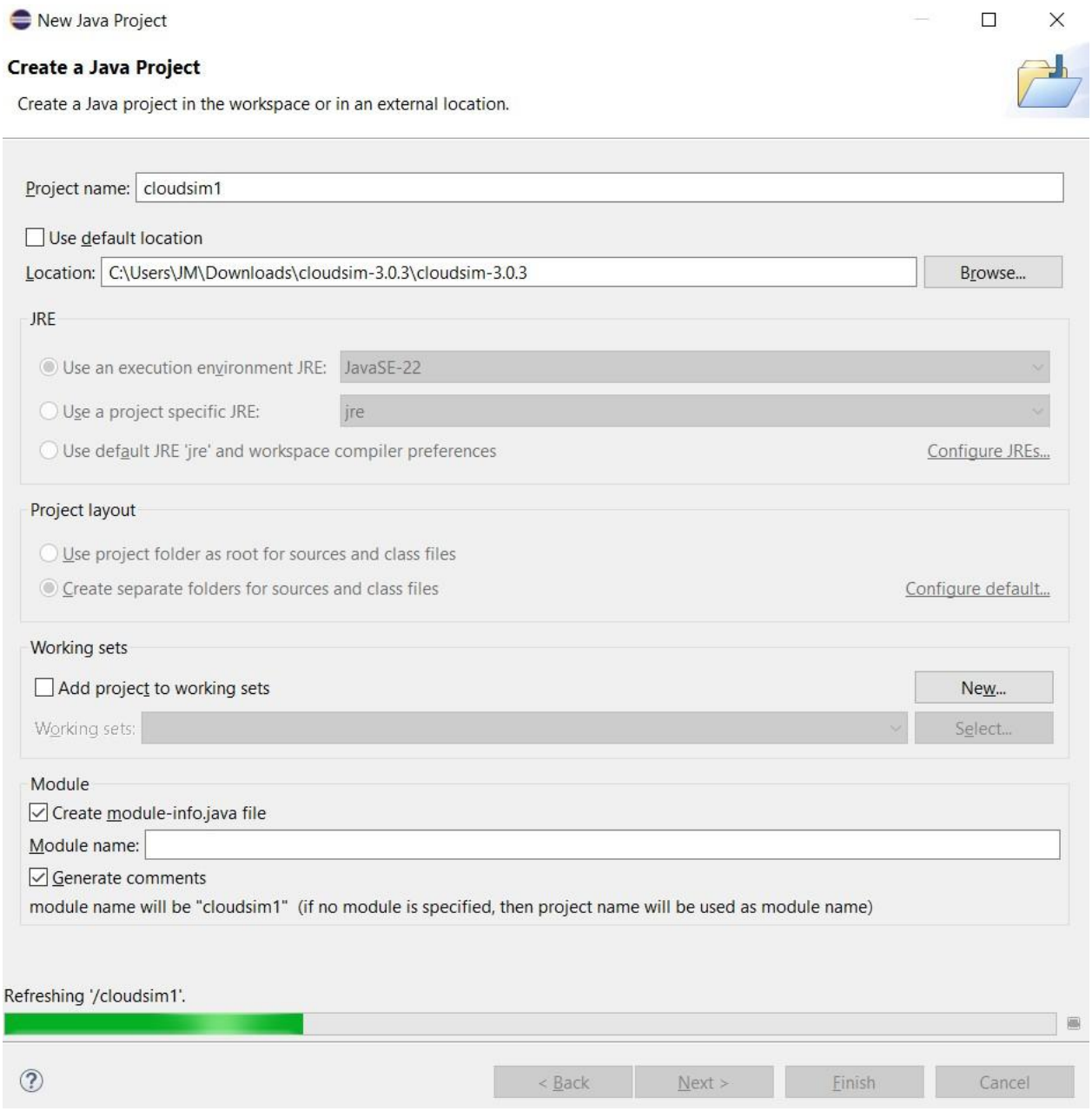


- Install Eclipse IDE

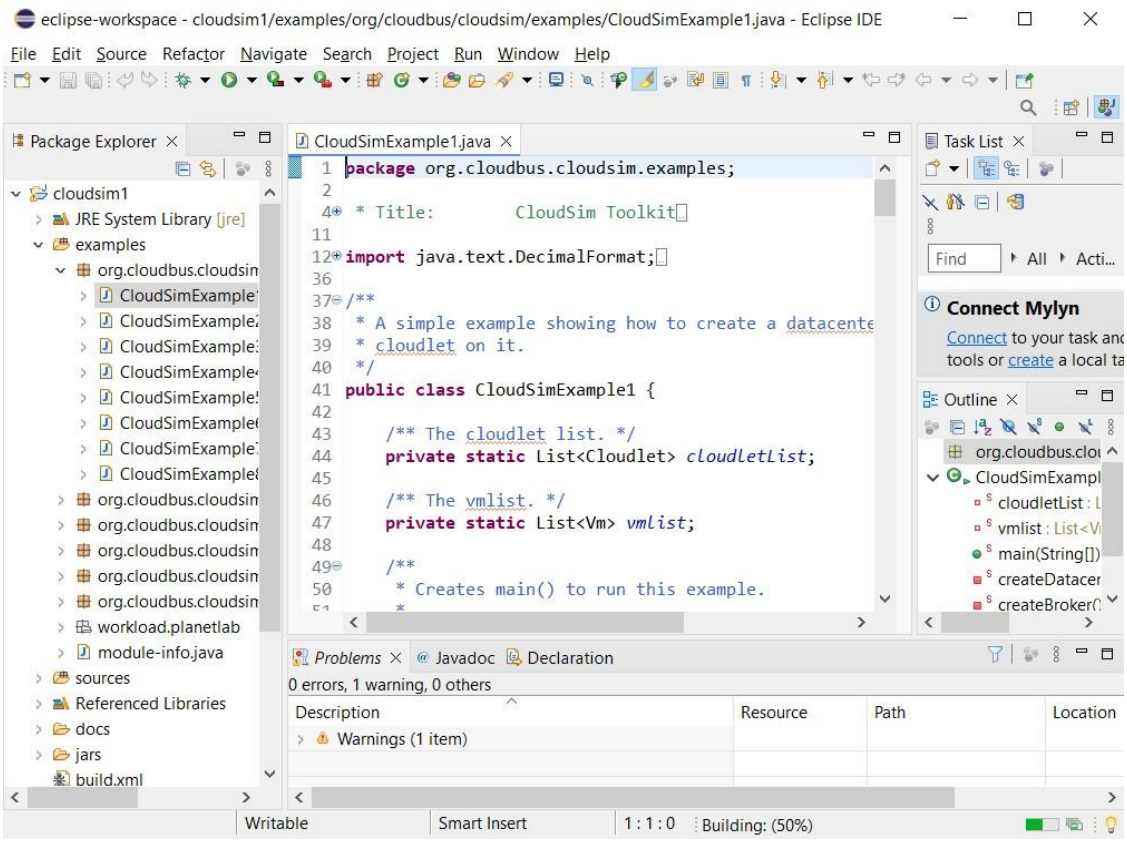


4. Run CloudSim in Eclipse

- Put up commons-math jar file into jar folder of Cloudsim3.0.3
- Build a new java project with Cloudsim3.0.3 folder Open Eclipse and create new java project

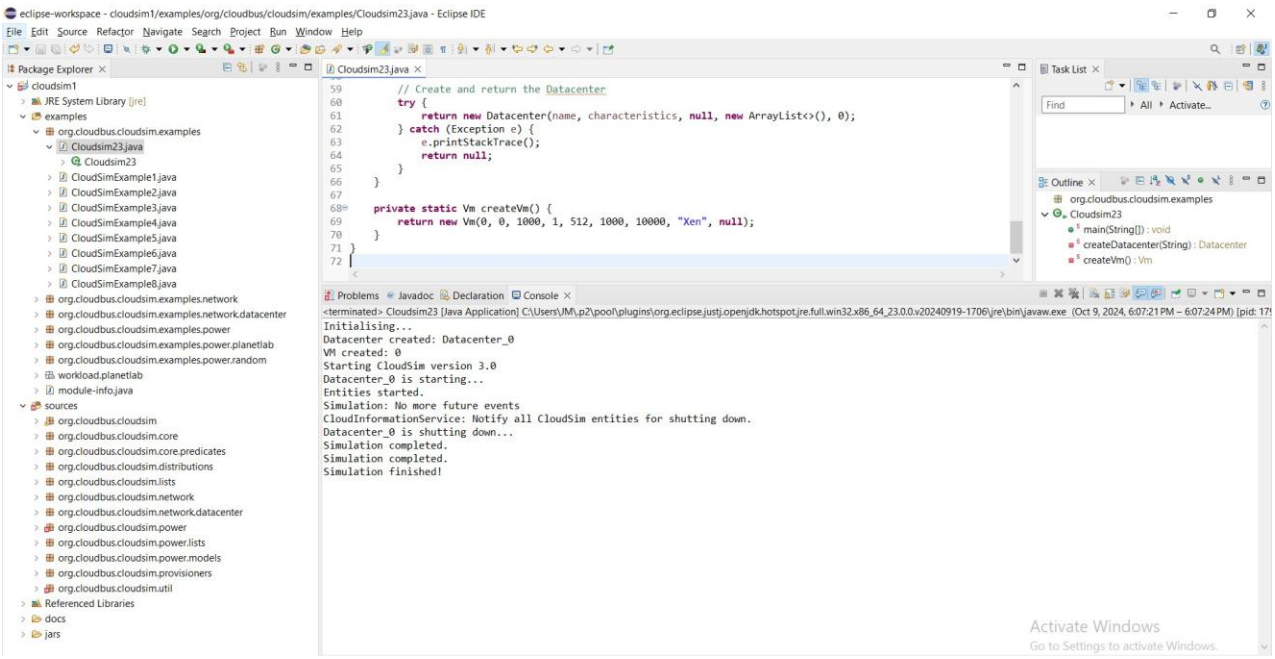


After creation of our project execute any example from the cloudsim examples



example CloudSim program: Creating a Datacenter, VM and Simulation

```
package org.cloudbus.cloudsim.examples;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.Vm; public
class SimpleVmCreation {     public static void
main(String[] args) {
      try {
         // Initialize CloudSim
         CloudSim.init(1, null, false);
         // Create a VM
         Vm vm = createVm();
         // Print VM information       System.out.println("VM created: " + vm.getId());
```



Output:

```
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Edge0 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0 VM is created on 0
1 VM is created on 1
2 VM is created on 2
3 VM is created on 3
4 VM is created on 0
5 VM is created on 1
6 VM is created on 2 7 VM is created on 3
app0
app1
app2
app3
app4
app5
app6
app7
app8
app9
app10
app11
app12
app13
```

```
Simulati
on: No
more
future
events
CloudInf
ormation
Service:
Notify
all
CloudSim
entities
for
shutting
down.
Datacenter_0 is shutting down...
Edge0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

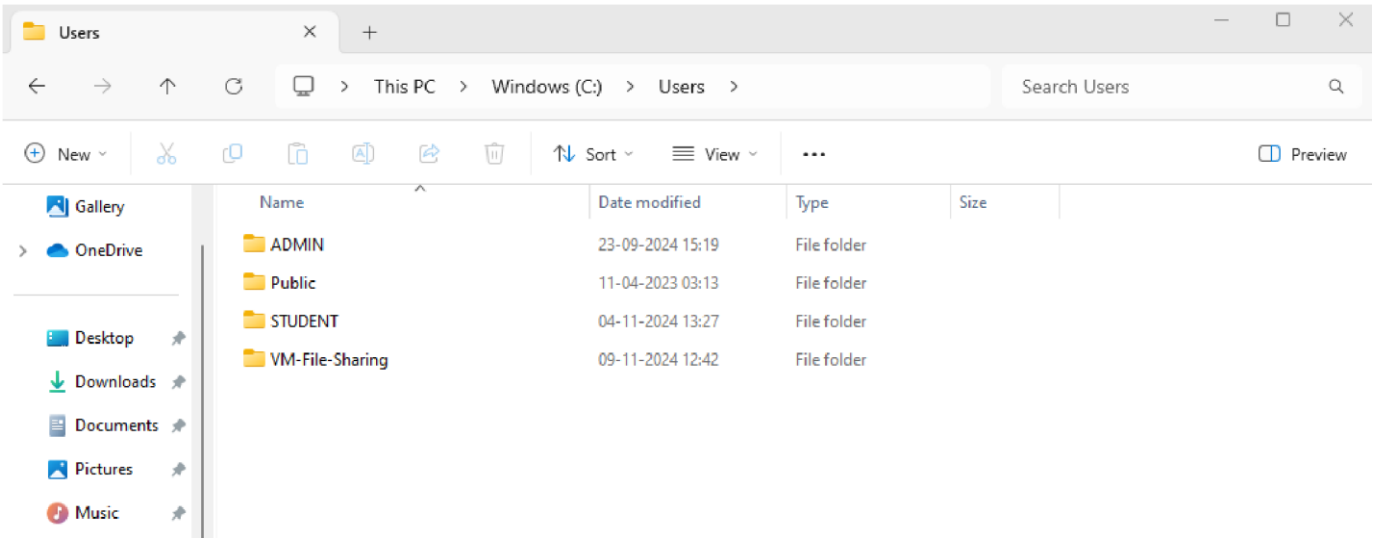
========== OUTPUT ==========
Cloudlet ID STATUS Data center ID VM ID Time Start Time Finish Time
1 SUCCESS 2 0 800 0 800
4 SUCCESS 2 0 800 0 800
7 SUCCESS 2 0 800 0 800
10 SUCCESS 2 0 800 0 800
0 SUCCESS 2 5 800 0 800
3 SUCCESS 2 5 800 0 800
6 SUCCESS 2 5 800 0 800
9 SUCCESS 2 5 800 0 800
13 SUCCESS 2 0 801 800 1601
numberofcloudlet 13 Cached 0 Data transfered 200000
CloudSimExample1 finished!
```

Week-9

Aim: Find a procedure to transfer the files from one virtual machine to another virtual machine.

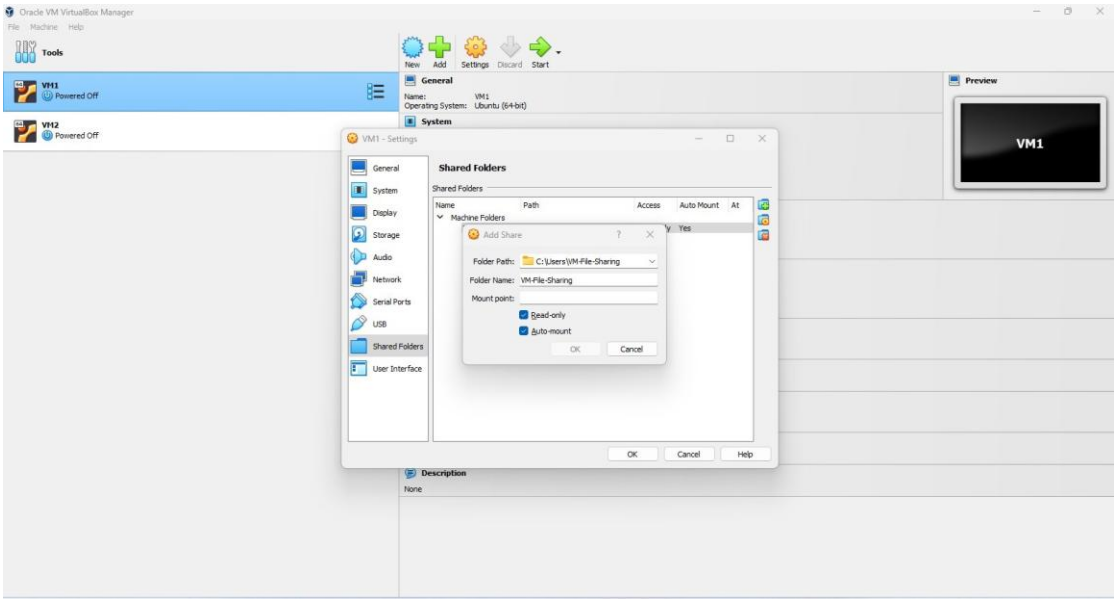
To set up file sharing between your Ubuntu VMs (VM1 and VM2) and the shared folder on your Windows host, follow these steps:

1. First create a Shared folder in C drive



2. Prepare the Shared Folder in VirtualBox Settings:

- Shut down VM1 and VM2 if they are running.
- In VirtualBox, select VM1, then go to **Settings > Shared Folders.**
- Click the **+** icon to add a shared folder.
- For **Folder Path**, select **C:\Users\VM-File-Sharing** on your Windows host.
- Set **Folder Name** (e.g., VM-File-Sharing).
- Check both **Auto-mount** and **Make Permanent** to ensure the folder is accessible each time.



➢Repeat these steps for VM2, adding the same shared folder.

3. Install Guest Additions in Both VMs:

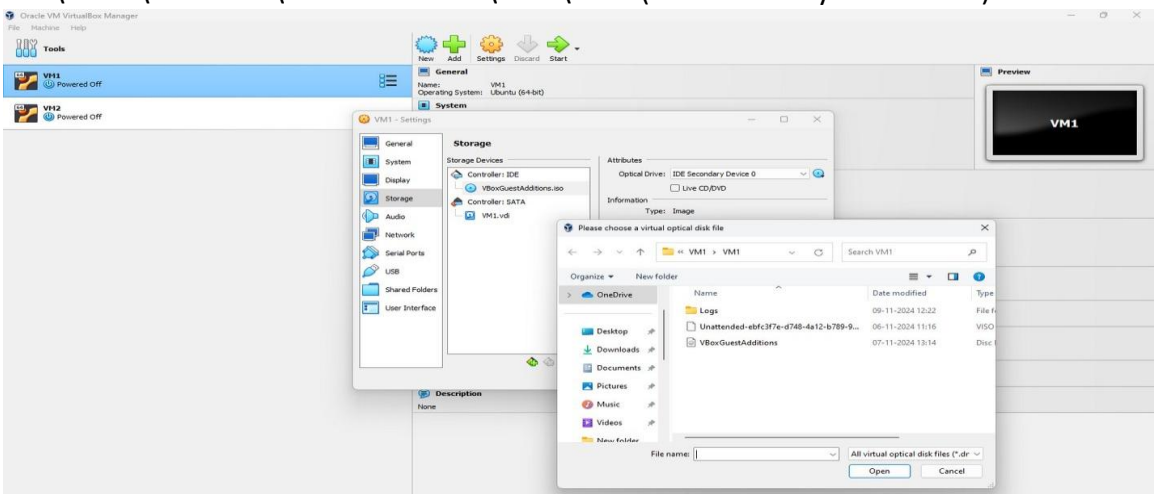
➢Start each VM (VM1 and VM2) one at a time.

➢Open VirtualBox, select VM1 / VM2, and go to Settings.

➢Under Storage, make sure that the VBoxGuestAdditions.iso is attached as the optical disk.

➢In the Storage settings, find Controller: IDE (or similar).

➢You should see VBoxGuestAdditions.iso listed under the optical drive. If it says Empty instead, click it, then select Choose a disk file… and locate VBoxGuestAdditions.iso in C:\Users\STUDENT\VirtualBox VMs\VM2\VM2 (or wherever you stored it).



➢Once the ISO is selected, click OK to save the changes.Open a terminal in the VM, then enter these commands to install Guest Additions:

sudo apt update

sudo apt install build-essential dkms linux-headers-$(uname -r)

(This command installs necessary packages for compiling and building software (especially kernel modules like the Guest Additions for VirtualBox).

sudo /media/$USER/VBox_GAs*/VBoxLinuxAdditions.run

(This command installs the VirtualBox Guest Additions, which are required to improve performance and enable features like file sharing and clipboard integration between the host and the VM.)

➢ Reboot each VM after installation.

4. Access the Shared Folder in Each VM:

- ➢ After rebooting, open a terminal in each VM.
- ➢ Add the VM user to the `vboxsf` group to access the shared folder:
  sudo usermod -aG vboxsf $USER
- ➢ Log out and log back in for the change to take effect.

5. Locate and Use the Shared Folder:

- ➢ To access it, navigate with: cd
  /media/sf_VM-File_sharing
- ➢ You can now transfer files by placing them in `C:\Users\VM-File_sharing` on Windows, which will automatically be accessible in the `/media/sf_VM-File_sharing` directory on each VM.

